GRADE 9 AGRICULTURE
CONTINUOUS ASSESSMENT TEST 2

TERM 1 | TIME: 1 HOUR 15 MINUTES | TOTAL: 50 MARKS

Name: ____ Adm. No.: _______________

School: _______________________________________ Date: ____________________

INSTRUCTIONS

- Answer all questions.
- Show clear steps where calculations are required.
- Use labelled sketches where appropriate.
- Observe safety when describing farm operations and handling materials.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Define the term crop rotation. (2 marks)

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2. State TWO advantages of using certified seeds. (2 marks)

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3. Differentiate between manure and compost. (2 marks)

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4. Give TWO reasons for thinning seedlings in a nursery or crop field. (2 marks)

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5. State TWO conditions necessary for successful seed germination. (2 marks)

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6. Name TWO methods of applying water to crops. (2 marks)

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7. State TWO signs of water stress in crop plants. (2 marks)

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8. Give TWO examples of beneficial organisms found in agricultural soils. (2 marks)

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9. State TWO reasons for harvesting crops at the correct stage of maturity. (2 marks)

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10. Give TWO precautions to observe when handling or applying farm chemicals. (2 marks)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Soil fertility and nutrient management (8 marks)

(a) Explain THREE ways of maintaining soil fertility on a small-scale farm. (3 marks)

(b) Describe TWO differences between organic and inorganic fertilisers. (4 marks)

(c) State ONE reason why fertilisers should be applied at the recommended rate. (1 mark)

12. Seed germination and crop establishment (8 marks)

(a) Describe FOUR conditions required for successful seed germination. (4 marks)

(b) Explain TWO benefits of raising seedlings in a nursery before transplanting. (2 marks)

(c) State TWO practices that reduce transplanting shock. (2 marks)

13. Water management and crop protection (7 marks)

(a) Explain THREE benefits of proper water management in crop production. (3 marks)

(b) Describe TWO cultural methods of controlling crop pests or diseases. (2 marks)

(c) State TWO reasons for removing diseased crop residues from a field. (2 marks)

14. Harvesting, storage and farm records (7 marks)

(a) State THREE indicators that may be used to determine crop maturity. (3 marks)

(b) Explain TWO measures that reduce post-harvest losses during storage. (2 marks)

(c) State TWO uses of farm records in managing a crop enterprise. (2 marks)

MARKING SCHEME – GRADE 9 AGRICULTURE CAT 2

Total: 50 marks

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. The practice of growing different crops on the same piece of land in a planned sequence over successive seasons. (2)

2. Any TWO: high germination percentage; true to variety; known quality; likely to be free from weed seeds and seed-borne diseases; uniform crop establishment. (1 each)

3. Manure is animal dung and/or urine, often mixed with bedding material; compost is decomposed organic matter made from plant and other biodegradable wastes. (1 each)

4. Any TWO: reduce competition for light, water and nutrients; allow healthy seedlings to develop; improve spacing and air circulation; remove weak or overcrowded seedlings. (1 each)

5. Any TWO: adequate moisture; oxygen/air; suitable temperature; viable seed. (1 each)

6. Any TWO: watering can; hose/sprinkler irrigation; drip irrigation; furrow irrigation; basin irrigation; flooding where appropriate. (1 each)

7. Any TWO: wilting; leaf rolling; yellowing or drying; stunted growth; premature leaf fall; poor flowering or fruit development. (1 each)

8. Any TWO: earthworms; nitrogen-fixing bacteria; decomposer fungi; beneficial soil bacteria; mycorrhizal fungi. (1 each)

9. Any TWO: improves quality and taste; reduces spoilage; ensures good market value; supports seed viability where seed is required; reduces field losses; allows proper storage. (1 each)

10. Any TWO: read and follow label instructions; wear recommended PPE; avoid eating or drinking while handling; keep chemicals away from children, food and water; avoid spraying in strong wind; wash after handling; use correct dosage. (1 each)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Soil fertility and nutrient management

(a) Any THREE, 1 mark each: apply well-prepared manure/compost; practise crop rotation with legumes; use mulching/cover crops; control soil erosion; practise agroforestry; apply fertilisers based on crop needs and recommendations.

(b) Any TWO differences explained, 2 marks each: organic fertilisers come from plant/animal materials while inorganic fertilisers are manufactured/mineral-based; organic fertilisers usually release nutrients more slowly while many inorganic fertilisers release readily available nutrients; organic materials add organic matter while inorganic fertilisers generally do not add significant organic matter. (4)

(c) Excess application can damage crops, cause nutrient imbalance, waste money or pollute water; under-application may not meet crop needs. (1)

12. Seed germination and crop establishment

(a) Any FOUR: moisture/water activates the seed and supports metabolic processes; oxygen is needed for respiration; suitable temperature supports enzyme activity; a viable, healthy seed is required. (1 each)

(b) Any TWO explained: easier care and watering; efficient use of seed; selection of healthy seedlings; protection from harsh conditions; allows early crop establishment while the main field is prepared. (1 each)

(c) Any TWO: transplant during cool hours or on a cloudy day; water seedlings before and after transplanting; minimise root damage; handle seedlings carefully; provide temporary shade where suitable. (1 each)

13. Water management and crop protection

(a) Any THREE: supplies water for photosynthesis and growth; prevents water stress; avoids waterlogging and root damage; improves nutrient uptake; supports uniform crop development; conserves water and reduces costs. (1 each)

(b) Any TWO described: crop rotation interrupts pest/disease life cycles; field sanitation removes breeding/infection sources; timely planting avoids peak pest periods; use of resistant varieties reduces disease damage; proper spacing improves airflow. (1 each)

(c) Any TWO: prevents spread of pathogens; removes pest breeding sites; reduces reinfection of healthy crops; improves field hygiene. (1 each)

14. Harvesting, storage and farm records

(a) Any THREE: change in colour; suitable size; firmness or dryness; pods/cobs reaching maturity; leaves or stems drying in crops where this is a maturity sign; days to maturity as indicated for the crop. (1 each)

(b) Any TWO explained: dry produce to safe moisture levels; store in clean, dry and ventilated facilities; protect from pests and rodents; use clean containers; sort and remove damaged produce; stack bags safely and keep them off damp floors. (1 each)

(c) Any TWO: monitor costs and income; plan inputs and labour; track yields and sales; assess profit or loss; support decision-making; provide evidence for future planning. (1 each)

Teacher's note: This is original teacher-prepared assessment material. Confirm that the relevant Grade 9 Agriculture content has been taught before administering. Accept other agriculturally correct answers and award marks for valid explanations.